Algorithms

Surface Blasting Design Calculator

1. Ask the user for Blasting Diameter (BD)
2. Ask the user for the average in hole density (ρ)
3. Ask the user for the Spacing to burden ratio (a)
4. Ask the user for Bench Height (BH)
5. Ask the user for Sub-drill (SD)
6. Ask the user for the Technical Powder Factor
7. Calculate linear charge mass = $M_c = \frac{\rho \times D^2}{1273}$
8. Calculate Stemming length (SL) = $20 \times \frac{Diameter(mm)}{1000}$
9. Calculate Burden = $\sqrt{\frac{Mc \times (BH - SL - SD)}{a \times PF_{tech} \times BH}}$
10. Calculate Actual Powder Factor = $\frac{(BH - SL \pm SD) \times Mc}{B \times S \times H}$

Fragmentation Calculator

1. Ask the user for the average in hole density (ρ)
2. Ask the user for Bench Height (BH)
3. Ask the user for the Relative Energy of Explosive (E)
4. Ask the user for Stemming Length (SL)
5. Ask the user for rock factor (A)
6. Ask the user for the powder factor (K)
7. Ask the user for the mass of explosive per hole (Q)
8. Calculate the mean fragmentation size (X_{50})
9. $X_{50} = AK^{-0.8} \times Q^{\frac{1}{6}} \left(\frac{115}{E}\right)^{\frac{19}{20}}$